Part D: The Cryo-Cooler

1. What state variables change?

   (a) In order to force the gas through the plug, which offers up considerable viscous friction, $P_1 > P_2$; it is this pressure difference that is the source of the force.

   (b) Viscous friction is dissipative, and the energy of moving through the plug is not allowed to go anywhere else, so the internal energy of the gas must **increase** as it moves through the plug, and then $U_1 < U_2$.

   (c) Though no heat is gained or lost, this is not a constant entropy process; that can be seen because it is an irreversible process. As such, $S_1 < S_2$

   (d) Since the process of moving across a pressure gradient imparts kinetic energy to an object, it is expected that the fluid velocity on the right will be higher than the left; in fact, we are told this is marginally true. Since mass is conserved, the volume of a mole of gas on the right must also be higher than the volume of a mole on the left, and $V_1 < V_2$.

   (e) The correct answer is $T_1 ? T_2$. If this were an ideal gas, one might expect $T_1 \leq T_2$ since $U \propto T$. But this is not an ideal gas, and U will be a function of temperature and density. As such, it is not possible to know the comparative relation between T_1 and T_2. That's the whole point of this problem, and the challenge of trying to make liquid helium.

   Marking scheme:

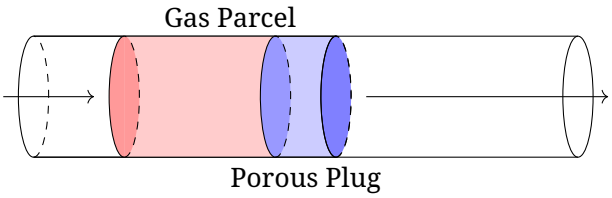
For each correct response	+0.2 pts
sum	**1.0 pt**

   Explanations by the students are not needed.

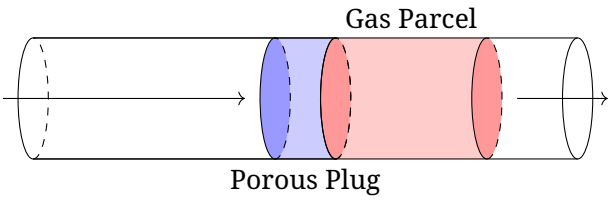
2. A mole of gas at P_1, V_1, T_1, U_1 enters the porous plug from the left, and that mole of gas exits the porous plug on the other side at P_2, V_2, T_2, U_2.

   Consider first a control volume approach

   The figure below shows the motion of a mole of gas through the plug; the mole is shown in pink. Gas to the left of the mole pushes the mole through the plug with a constant force $P_1 A$ through a volume V_1.



   The mole of gas moves through the plug to the right hand side, in the process pushing on the air to the right of the mole with a constant force $P_2 A$, through a volume V_2.



   The work that the surrounding gas in region 1 does on the gas pushing it into the plug is

$$W_1 = P_1 V_1$$

   because the pressure is constant, and the effective change of volume is V_1. Similarly, when the gas enters region 2 it must displace a volume V_2 of gas that was already there, so

$$W_2 = -P_2 V_2$$

   The net work is then

$$W_{net} = P_1 V_1 - P_2 V_2 \tag{47}$$

   Since there is no heat exchanged,

$$U_2 - U_1 = \Delta U = Q + W_{net} = P_1 V_1 - P_2 V_2 \tag{48}$$

   which implies

$$\Delta U = U_2 - U_1 = P_1 V_1 - P_2 V_2.$$

   Upon rearranging

$$U_2 + P_2 V_2 = U_1 + P_1 V_1$$

   and therefore

$$U + PV$$

   is a conserved quantity.

   Marking scheme:

Compute correct W_1	0.1 pts
Compute correct W_2	0.1 pts
Write energy law, Eq 48	0.2 pts
Show $U + PV$ conserved	0.2 pts
sum	**0.6 pt**

   Consider instead a differential approach

   Another way to look at this problem is to focus on a differential sample of gas as it moves through the plug.

   The figure below illustrates this

   The total energy of parcel of molar size δm has two relevant energy terms: the internal energy δU and the bulk kinetic energy δK. It has a volume δV. These four quantities are extrinsic, but to simplify notation, we will drop the δ. It's still there, just invisible.

   For simplicity's sake, assume a cylindrical shape to the parcel, with an end cap area δA and a length dx. Once again, we will drop the δ. There are three forces that act on the shape, one associated with pressure on the left end, one associated with pressure on the right end, and frictional force associated with viscosity against the walls of the container.

Since this is a parcel of differential length dx, the net force associated with the pressure difference between the ends is

$$F_{ends} = -V\frac{dP}{dx}$$

where V is again the volume of the cylinder.

But this force is (mostly) balanced by the viscous frictional force F_{walls} with the walls of the sponge; these two forces effectively add to zero. In fact, it is the viscous forces with the wall that cause the pressure gradient across the sponge.

The bulk kinetic energy of the parcel does not change significantly as it moves through the sponge. This is seen in that the bulk speed of the gas doesn't change significantly as it moves through the sponge.

The problem with this approach is that the system is not in thermodynamic equilibrium; the process is not reversible, so it is not possible to attach well defined state variables. This means that

$$dU = TdS - PdV \tag{49}$$

is not a function that can be integrated; in fact, $dS \neq 0$ from the previous part of the problem. Arguing that $VdP = -TdS$ is rather handwavy, and resolving this actually requires considering a control volume approach.

Still, the energy conservation ideas still hold true, even if thermodynamically poorly defined, so

$$dU = -PdV - VdP$$

since the part associated with $-VdP$ doesn't change the bulk kinetic energy, and instead dissipates into internal energy of the gas.

The result is that

$$dU = -d(PV)$$

or

$$U + PV$$

is a constant

Marking scheme:

Traditional $\delta W = -PdV$	0.1 pts
Bulk kinetic $\delta K = -VdP$	0.1 pts
Explain where δK goes	0.1 pts
Differential Eq 49	0.1 pts
integrate $U + PV$ constant	0.1 pts
sum	**0.5/0.6 pt**

Because of the many subtle traps, this approach will not get the same number of points as the control volume approach.

Writing $dU = -PdV$ and integrating to find $U + PV$ is constant gets only 0.2 pts. This is because there are several errors: the differential is poorly defined within the sponge; because the state variables are poorly defined; P is not a constant; so you can't actually integrate it; and the work done in this case is not correctly computed. Four wrongs don't make a right.

3. One can find pressure on this graph by applying

$$dU = TdS - PdV$$

and then requiring constant entropy so that $dS = 0$, and then

$$P = -\left(\frac{\partial U}{\partial V}\right)_S \tag{50}$$

which are the negative slopes of the constant entropy curves on a $U - V$ graph.

Another approach to find pressure is to consider a line of constant U, then

$$\frac{P}{T} = \left(\frac{\partial S}{\partial V}\right)_U$$

Then

$$U + PV = U - \left(\frac{\partial U}{\partial V}\right)_S V$$

is the conserved quantity.

Now $-(\partial U/\partial V)_S$ is measured only at the point V_1, U_1, and is the slope of the tangent line to the constant entropy curve. Following that tangent line back a distance V takes it to an intercept with the U axis, and that intercept is then the conserved quantity.

More mathematically, define a function H

$$H = U + PV$$

then

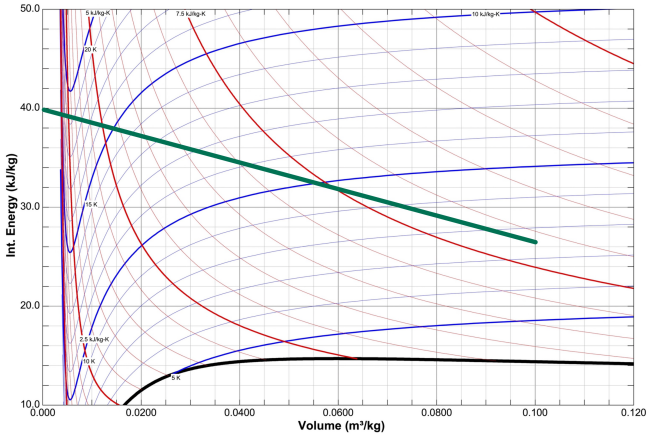
$$U = H_2 - P_2 V$$

is the equation of a line,

$$U = H_2 + \left(\frac{\partial U}{\partial V}\right)_S\Big|_2 V \tag{51}$$

with the U intercept equal to the conserved H_2.

An estimate can be made visually, but it is difficult to be accurate. Try constructing a line from the point $V_2 = 0.120$, $T_2 = 7.5$ that is tangent to the local isentrope, and the result will intercept the U axis. This result is somewhere around 40. This is shown in green below.



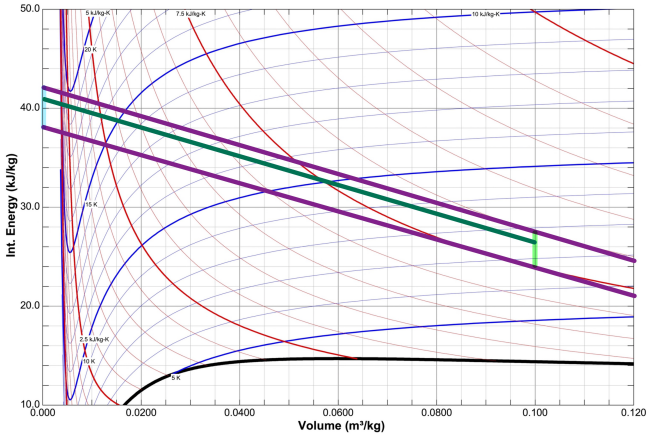
Now to improve the result.

Draw a line out from 39 that is tangent to the nearest isentrope to $V_2 = 0.100$, $T_2 = 7.5$; draw another

line out from 41 that is also tangent to the nearest isentrope to $V_2 = 0.100$, $T_2 = 7.5$. These are shown in purple below.

Measuring the distance with a ruler, find the fractional distance between the two purple lines to the point $V_2 = 0.100$, $T_2 = 7.5$ along the highlighted green line. It is about 75% the way from the bottom purple line. This means that the conserved quantity ought be 75% the way up on the highlighted blue section on the graph. A line connecting the two is shown in green.

This point is about 41 kJ/kg. The actual value for the conserved quantity is $U + PV = 40.7$ kJ/kg.



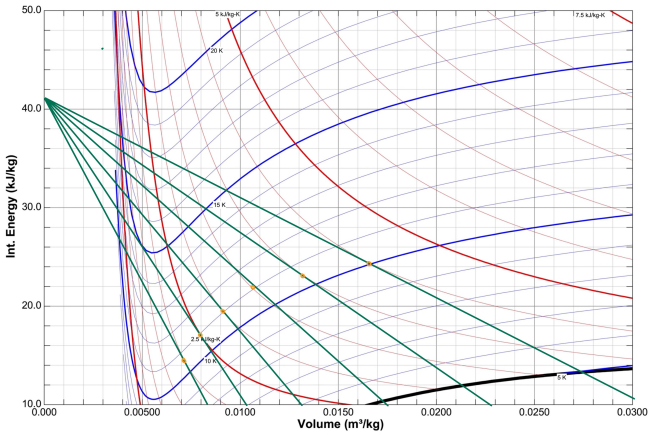
Marking scheme:

Pressure formula stated, Eq 50	0.2 pts
Tangent intercept concept	0.4 pts
A first estimate for H	0.2 pts
Upper bound for estimate set	0.2 pts
Lower bound for estimate set	0.2 pts
Interpolated estimate set	0.2 pts
$40.5 < H < 41.0$	0.2/0.2 pts
$40.2 < H < 41.2$	0.1/0.2 pts
sum	**1.4 pt**

As the task asks for a graphical construction, and it is not possible to construct an accurate tangent to the isentrope at $T_2 = 7.5$K based on a single line, students *must* do something to improve or verify the result, even if it is correct on the first guess. Hence the upper and lower bound approach and interpolation, or something equivalent.

4. Draw a series of radial lines out from the conserved point that are tangent to lines of constant entropy. Mark the tangent point. Connect with a smooth curve; this curve is the set of points U_1 as a function of V_1 that has the conserved quantity. Look for the maximum temperature intercept.

   This happens at about $T_1 = 11$K. If T_1 is higher than this, it would not be possible to cool down to $T_2 = 7.5$K.



Students don't need to draw every line, as with a straight edge one can find the tangent that maximizes the temperature T_1 by shifting it around visually.

Line starts from student's H	0.2 pts
Line intercepts an isentrope	0.2 pts
The isentrope matches max T_1	0.2 pts
Stated T_1 within 0.5K of student's construction	0.1 pts
$10\text{K} \leq T_1 \leq 12\text{K}$	0.1 pts
sum	**0.8 pt**

5. Using the slope of the line from the conserved quantity to the maximum temperature point, compute the pressure.

   Using the results from above,

$$P_1 = -\frac{(41) - (10)}{(0) - (0.0170)} = 1.8\,\text{MPa}$$

If they didn't know to use slope by this point, they can't generate an answer. As such, they would already have received points for the pressure formula, and we only consider the numerical result

P agrees with the slope of the graph	0.1 pts
$1.6\,\text{MPa} \leq P_1 \leq 2.4\,\text{MPa}$	0.1 pts
sum	**0.2 pt**

T3: Scaling laws (8 pts)

Note: A correct numerical answer provided with at least two significant figures receives full marks. Inappropriate use of equality will lead to a penalty of 0.1 pts for each part of the question.

Task A: Spaghetti (2 pts)

This is section 2.2.2 (Statics) of the syllabus.

Consider only the left half of the spaghetti straw.

Torque balance at its right endpoint implies that the torque applied to its right endpoint must balance out the torque due to gravity: $\tau \propto ml \propto d^2l^2$. This torque arises from the gradient in the horizontal stress. If the typical horizontal stress is σ, then the typical force is $F \propto \sigma d^2$, so the torque is $\tau \propto Fd \propto \sigma d^3$. Hence, we obtain

$$\sigma d^3 \propto d^2 l^2 \implies l \propto \sqrt{d},$$

so

$$l' = \sqrt{\frac{d'}{d}} l = \sqrt{10} \cdot 50\,\text{cm} = 158\,\text{cm}.$$

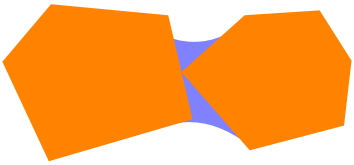
Marking scheme:

$\tau \propto d^2l^2$	0.4 pts
$F \propto \sigma d^2$	0.5 pts
$\tau \propto \sigma d^3$	0.5 pts
$l \propto \sqrt{d}$	0.4 pts
Answer: $158\,\text{cm}$	0.2 pts

Task B: Sand castle (2 pts)

This is Section 2.2.2 (statics) and 2.2.5 (hydrodynamics) of the syllabus

Due to wetting of the surfaces of the sand grains and its large surface tension water acts like a glue for sand. This means that all the grains need to be bound together by air-water interface. To achieve this there needs to be neither too little nor too much water: if there is too little water, most of the grains are dry with no surface tension binding them, and if there is too much water, almost all the grains are immersed into water, and again, there is no surface tension binding the grains. So, the overall strength of the buildings from wet sand depends on the water content; we assume that for the both types of sand, the water content is optimal, and the shape of the grains is statistically similar. Let us consider two neighbouring grains connected by a water meniscus — or "neck", as we shall be referring to it henceforth. Note that the "neck" may extend perpendicularly to the figure plane far away; so, more specifically, what the word "neck" will refer to is that part of the water-air interface for which the closest two grains are the ones under consideration.



There are two processes binding the sand grains together. The first one is the force due to the surface tension, $F_1 = \gamma l$, where γ denotes the surface tension coefficient, and l — the perimeter of the "neck"; with $l \sim r_g$, where r_g denotes the length scale of a single grain, we obtain $F_s \sim \gamma r$. The second one is the pressure force caused by the negative capillary pressure in the neck, $F_p = \Delta p A$, where A is the cross-sectional area of the "neck", and $\Delta p \sim \gamma / r$. With $A \sim r^2$ we obtain $F_2 \sim \gamma r$. Thus, the both components are of the same order of magnitude and using either of them will lead to the correct scaling law. These forces press the grains against each other, hence the normal force and friction force between the grains is also on the order of F_s and F_p.

Solution 1:

Based on what has been said above, the typical force needed to delocate a grain of sand is $F_g \propto r_g$. The force needed to delocate an entire horizontal layer of sand is then $\propto F_g N_l$, where $N_l \sim A/r_g^2$ is the number of grains in a layer. The force of cylinder destruction F thus satisfies

$$F \propto F_g N_l \propto r_g / r_g^2 = r_g^{-1} \propto V_g^{-1/3},$$

so

$$F_{fg} = (1/10)^{-1/3} \cdot F_{cg} = 21.5\,\text{N}.$$

Marking scheme:

a) $F_s \propto r_g$ and $F_p \propto r_g$	0.5 pts
one of the two missing	-0.1 pts
b) $F_g \propto r_g$	0.5 pts
c) $F \propto F_g N_l$	0.5 pts
d) $F \propto r_g^{-1}$	0.3 pts
Answer: $21.5\,\text{N}$	0.2 pts

Notes: If the student only qualitatively explains the mechanism by which the grains of sand are held together, a maximum of 0.5 pts are given. Points b)-c) are given only if derived from a).

Solution 2:

We have seen above that grains to one side of a fictitious surface exert force per cross-sectional area on the order of magnitude as the capillary pressure $\Delta p \sim \gamma / r_g$. In order to get the grains moving, a pressure of the same order of magnitude needs to be applied externally. For both cylinders, the surface area where the force is applied is the same, hence the force scales as the capillary pressure, $F \propto 1/r_g \propto V^{-1/3}$.

Marking scheme:

The applied pressure must be $\sim \Delta p$	0.6 pts
The curvature radius of the interface is $\sim r_g$	0.6 pts
Capillary pressure $\Delta p \sim \gamma / r_g$	0.6 pts
Answer: $21.5\,\text{N}$	0.2 pts

Note: If the contribution of surface tension is neglected, 0.1 pts are subtracted.

Solution 3:

The compression force serves to break the surface tension bonds between sand grains.

Consider the energy E required to push a single layer of sand into the layer beneath it. $E \propto F r_g$, where F is the

force required and r_g is the typical height of a layer (i.e., the typical length scale of a grain).

On the other hand, $E = \gamma \Delta A$, where γ is the surface tension of water and ΔA is the total amount by which the surface of the water in the layer stretches before all the "water bonds" between the sand grains are broken.

Here, ΔA is proportional to the area A of a layer and is thus a constant between the two cylinders. Hence, $E \propto F r_g$ is a constant between the two cylinders, i.e., $F \propto r_g^{-1}$.

Marking scheme:

$E \propto F r_g$	0.5 pts
$E \propto \gamma \Delta A$	0.5 pts
$\Delta A \propto A$	0.5 pts
$F \propto r_g^{-1}$	0.3 pts
Answer: $21.5\,\mathrm{N}$	0.2 pts

Note: If the student only qualitatively explains the mechanism by which the grains of sand are held together, a maximum of 0.5 pts are given.

Solution 4:

First of all, the force F should be proportional to the cylinder's base area A. The force required to destroy a cylinder with base area $A = nA_0$ is equal to the force required to destroy n cylinders each with base area A_0. As a result, $F \propto n \propto A$.

In addition, F depends on the grain's length scale r_g and the water's surface tension γ. Dimensional analysis thus gives

$$F \propto \frac{A\gamma}{r_g} \propto r_g^{-1}$$

for fixed A and γ.

Marking scheme:

$F \propto A$	0.6 pts
$F = F(A, r_g, \gamma)$	0.6 pts
$F \propto \frac{A\gamma}{r_g}$	0.6 pts
Answer: $21.5\,\mathrm{N}$	0.2 pts

Note: If the student only qualitatively explains the mechanism by which the grains of sand are held together, a maximum of 0.5 pts are given.

Task C: Interstellar travel (2 pts)

This is Section 2.5 (Relativity) of the syllabus

Let $T = 50\,\mathrm{yrs}$ be the astronauts' total travel time. For maximal travel distance, the spaceship accelerates at constant proper acceleration $a = g$ for proper time $T/4$, during which a distance of d is traveled. The spaceship then decelerates at $a = -g$ for proper time $T/4$ to come to a rest, during which another distance d is traveled. The spaceship then returns to Earth using the same procedure.

Notes: Formula relating acceleration to proper acceleration is not considered as a basic SR formula and therefore if the formula is written without motivation, 0.2 pts are subtracted.

Solution 0: (incorrect)

If we ignore relativity, then $d \propto \frac{1}{2} g t^2 \propto g$, which gives an answer of 1.5.

Marking scheme:

$d \propto g t^2$	0.2 pts
Answer: 1.5	0.1 pts

Solution 1:

One way to approach the problem is to notice that constant acceleration in spaceship's frame means a constant force in the Earth's frame. This follows directly from the Lorentz transform for the electromagnetic field, more specifically from the fact that when going to a frame moving parallel to the x-axis, the x-directional electric field E_x remains unchanged. Hence, on the one hand, the force $F_x = eE_x$ exerted on an accelerating particle of rest mass m_0 and carrying a charge e remains constant in the lab frame. On the other hand, the acceleration of that particle in an inertial frame moving with velocity v, where v denotes the particle's velocity at a certain moment of time t, is always equal to eE_x/m_0, regardless of the value of t, i.e. constant in time.

Those who are not familiar with the Lorenz transform for electromagnetic field can derive the above described property from the Lorenz transform for momentum and coordinates. We use again (i) the lab frame, and (ii) an inertial frame moving with velocity v, where v denotes the spaceship's velocity at a certain moment of time which will be used as the origin, $t = t' = 0$; let primes denote quantities in the second frame. Assuming a very short time period t, we can neglect terms quadratic in time so that in the frame (ii), the momentum, coordinate and the relativistic mass can be expressed as $p' = F't'$, $x' = 0$, $m' = m_0$, respectively; applying the Lorenz transform yields $t = \gamma t'$ and $p = \gamma(F't' + m_0 v) = tF' + \gamma m_0 v$. On the other hand, in the frame (i), $p = \gamma m_0 v + Ft$; comparing this with the previous result yields $F = F'$.

It appears that in either case, the spaceship's speed will reach almost c much faster than the travel time. Hence, using for convenience the system of units where $c = 1$, the travel distance x equals with a very good precision the travel time t, $x = t$.

What is left to do is to relate t to the proper time τ,

$$\mathrm{d}\tau = \frac{\mathrm{d}t}{\gamma} = \mathrm{d}t \frac{m_0}{\sqrt{m_0^2 + m_0^2 g^2 t^2}};$$

upon integration we obtain

$$\tau = \mathrm{asinh}(gt)/g \Rightarrow x \approx t = \sinh(g\tau)/g \approx \exp(g\tau)/2g.$$

So we conclude that the ratio of the travel distances is

$$\frac{d_2}{d_1} = \frac{g}{1.5g} \exp(1.5g\tau - g\tau) = \frac{2}{3} \exp(gT/8) \approx 480.$$

Note that an exact relationship between x and t could have been obtained by expressing the energy of the spaceship as $m = m_0 + m_0 g x$, and the momentum as $p = m_0 g t$. Then the Lorenz invariant $(m_0 + m_0 g x)^2 - (m_0 g t)^2 = m_0^2$ yields $x(x + 2/g) = t^2 = \sinh^2(g\tau)/g^2$, hence $x = [\cosh(g\tau) - 1]/g$.

F_x is Lorentz invariant	0.4 pts
$x \approx t$	0.4 pts
$\mathrm{d}\tau = \frac{\mathrm{d}t}{\gamma}$	0.2 pts
$\gamma^{-1} = m_0/m$	0.2 pts
$m = \sqrt{m_0^2 + p^2}$	0.2 pts
$p = m_0 g t$	0.2 pts
$t = \sinh(g\tau)/g$	0.2 pts
Answer: 480	0.2 pts

Remark: if integration boundaries for distance or proper time are wrong by a factor of 0.5, 2, 4, etc., -0.1 pts.

Solution 2:

Let w be the rapidity of the spaceship, defined as $w \equiv \tanh^{-1}(\beta)$, where β is the spaceship's velocity. Then $\beta = \tanh w$, the Lorentz factor $\gamma = \cosh w$, and its momentum $p = m_0 \sinh w$.

As shown by Solution 1, a spaceship experiencing a constant proper acceleration g experiences a constant three-force

$$F = m_0 g = \frac{\mathrm{d}p}{\mathrm{d}t} = m_0 \cosh w \frac{\mathrm{d}w}{\mathrm{d}t} \implies \frac{\mathrm{d}w}{\mathrm{d}t} = \frac{g}{\cosh w}.$$

Meanwhile, time dilation relates t to the spaceship's proper time τ as

$$\frac{\mathrm{d}t}{\mathrm{d}\tau} = \gamma = \cosh w \implies \frac{\mathrm{d}w}{\mathrm{d}\tau} = \frac{\mathrm{d}w}{\mathrm{d}t}\frac{\mathrm{d}t}{\mathrm{d}\tau} = g.$$

Integrating yields $w = g\tau$. Recalling that $dt = \gamma d\tau$, we get the following as the total distance traveled over a quarter of the spaceship's trip:

$$\begin{aligned} d = \int_0^{T/4} \beta\gamma \,\mathrm{d}\tau &= \int_0^{T/4} \tanh w \cosh w \,\mathrm{d}\tau \\ &= \int_0^{T/4} \sinh g\tau \,\mathrm{d}\tau = \frac{1}{g}(\cosh(gT/4) - 1). \end{aligned}$$

The answer is thus

$$\begin{aligned} \frac{g_1}{g_2}\frac{\cosh(g_2 T/4c) - 1}{\cosh(g_1 T/4c) - 1} &= \frac{10}{15}\frac{\cosh(19.72) - 1}{\cosh(13.15) - 1} \\ &\approx \frac{2}{3} e^{19.72 - 13.15} = 480. \end{aligned}$$

Marking scheme:

$\frac{\mathrm{d}}{\mathrm{d}t}(m_0 \sinh w) = m_0 g$	0.5 pts
$\frac{\mathrm{d}w}{\mathrm{d}t} = \frac{g}{\cosh w}$	0.1 pts
$\frac{\mathrm{d}t}{\mathrm{d}\tau} = \cosh w$	0.4 pts
$\frac{dw}{d\tau} = g$	0.1 pts
$w = g\tau$	0.1 pts
$\frac{d}{2} = \int_0^{T/4} \beta\gamma \,\mathrm{d}\tau$	0.3 pts
$\frac{d}{2} = \int_0^{T/4} \tanh w \cosh w \, d\tau$	0.2 pts
$\frac{d}{2} = \frac{1}{g}(\cosh(gT/4) - 1)$	0.1 pts
Answer: 480	0.2 pts

Remark: if integration boundaries for distance or proper time are wrong by a factor of 0.5, 2, 4, etc., -0.1 pts.

Solution 3: The problem can be also solved by using the trick introduced in 1905 by Henri Poincaré [Poincaré, M.H. Sur la dynamique de l'électron. Rend. Circ. Matem. Palermo 21, 129–175 (1906)] of depicting things in $x - \mathrm{i}t$-diagram. The benefit of using this diagram is that the relativistic invariant $x^2 - t^2$ transforms into Euclidean squared distance $x^2 + \theta^2$ with $\theta = \mathrm{i}t$. This means that in that diagram, we can use the knowledge of Euclidean geometry. In particular, the Lorentz transform is now the rotation of the Euclidean $x - \mathrm{i}t$-space by an angle $\alpha = \arctan \frac{v}{\mathrm{i}c}$. Now, consider the trajectory of the space ship; its infinitesimal arc length is $\mathrm{i}c\mathrm{d}\tau$, where $\mathrm{d}\tau$ is the differential of the proper time, and the infintesimal rotation angle of its tangent is $\mathrm{d}\alpha = \arctan(\mathrm{d}v/\mathrm{i}c) = \mathrm{d}v/\mathrm{i}c = g\mathrm{d}\tau/\mathrm{i}c$. Therefore, the curvature radius $R = \mathrm{i}c\mathrm{d}\tau/\mathrm{d}\alpha = -c^2/g$ is constant, i.e. the trajectory is a circle of radius R. Now we can easily relate the travel distance x to the arc length $\mathrm{i}c\tau$:

$$x = R(1 - \cos\alpha) = R\left(1 - \cos\frac{\mathrm{i}c\tau}{R}\right) = \frac{c^2}{g}\left(\cosh\frac{g\tau}{c} - 1\right).$$

Marking scheme:

$R =$ const in x-ict-diag.	0.5 pts
$R = -g^2/c$	0.5 pts
missing '–'	-0.2 pts
partial credit for $R = \frac{\mathrm{i}c\tau}{\mathrm{d}\alpha}$	0.2 pts
$x = R(1 - \cos\alpha)$	0.5 pts
$\frac{c^2}{g}\left(\cosh\frac{g\tau}{c} - 1\right)$	0.3 pts
Answer: 480	0.2 pts

Solution 4: The problem can be solved by using the velocity addition formula. Let $v = \beta c$ be the speed of the spaceship in the lab frame, t be the lab time, and τ — the proper time. Also, we consider a frame which moves with constant speed v in which the spaceship accelerates from rest:

$$\beta + \mathrm{d}\beta = \frac{\beta + \frac{g\mathrm{d}\tau}{c}}{1 + \frac{\beta g\mathrm{d}\tau}{c}} = \beta + \frac{g\mathrm{d}\tau}{c}(1 - \beta^2).$$

Thus,

$$\frac{\mathrm{d}\beta}{1 - \beta^2} = \frac{g\mathrm{d}\tau}{c} \implies \beta = \tanh(\frac{g\tau}{c}).$$

From relativistic time dilation formula we obtain

$$\mathrm{d}t = \frac{\mathrm{d}\tau}{\sqrt{1 - \beta^2}} = \cosh(\frac{g\tau}{c})\mathrm{d}\tau$$

so that the travel distance

$$\frac{d}{2} = \int v\mathrm{d}t = c\int_0^{T/4} \sinh(\frac{g\tau}{c})\mathrm{d}\tau = \frac{c^2}{g}\left[\cosh\left(\frac{gT}{4c}\right) - 1\right]$$

which leads to the same answer as before.

a) $\beta + \mathrm{d}\beta = \frac{\beta + \frac{g\mathrm{d}\tau}{c}}{1 + \frac{\beta g\mathrm{d}\tau}{c}}$	0.3 pts
b) $\frac{\mathrm{d}\beta}{1-\beta^2} = \frac{g\mathrm{d}\tau}{c}$	0.2 pts
c) $\beta = \tanh(\frac{g\tau}{c})$	0.2 pts
d) $\mathrm{d}t = \frac{\mathrm{d}\tau}{\sqrt{1-\beta^2}}$	0.3 pts
e) $\mathrm{d}t = \cosh(\frac{g\tau}{c})\mathrm{d}\tau$	0.2 pts
f) $\frac{d}{2} = \int v\mathrm{d}t$	0.2 pts
g) $\frac{d}{2} = c\int_0^{T/4} \sinh(\frac{g\tau}{c})\mathrm{d}\tau$	0.2 pts
h) $\frac{d}{2} = \frac{c^2}{g}\left[\cosh\left(\frac{gT}{4c}\right) - 1\right]$	0.2 pts
i) Answer: 480	0.2 pts

Remark: if integration in f) is done over proper time, no points are given for f). If integration boundaries for distance or proper time are wrong by a factor of 0.5, 2, 4, etc., -0.1 pts.

Task D: That sinking feeling (2 pts)

(This is Sections 2.2.5 (Hydrodynamics) and 2.4.1 (Single oscillator) of the syllabus

Solution 1: The oscillation of the half-sunk sphere is driven by gravity. The non-damped angular frequency depends on the gravitational acceleration and a characteristic length, which is, for a sphere, its radius r, so

$$\omega_0 \propto \sqrt{g/r}$$

is the only dimensionally correct possible function.

The drag force F_d depends on the sphere's speed v [m/s], its size r [m], and viscosity of the liquid η [Pa·s]. Dimensional analysis thus gives $F_d \propto \eta r v$. The damping factor is thus

$$\beta = \frac{F_d}{2mv} \propto \frac{\eta r}{m}.$$

Since the mass scales with r^3, we have

$$\beta \propto \frac{1}{r^2}.$$

Then the relation

$$\frac{\beta^2}{\omega_0^2} = 1 - \frac{\omega^2}{\omega_0^2}$$

scales as

$$\frac{\beta^2}{\omega_0^2} \propto \frac{1}{r^3}$$

Oscillations only occur if $\beta/\omega_0 < 1$, so solve

$$\frac{r}{r_0} = \sqrt[3]{1-(0.99)^2} = 0.271$$

Notes:

1. To obtain $\omega_0 \propto 1/\sqrt{r}$ without dimensional analysis, note that a small displacement y changes the submerged volume of the ball by $\Delta V \propto r^2 y$, so the change in buoyant force $F \propto r^2 y$, which gives $\omega_0 = \sqrt{k/m} \propto \sqrt{r^2/r^3} = \sqrt{1/r}$.

2. To obtain $F_d \propto \eta r v$ without dimensional analysis, note that the typical length scale l in the variations in the velocity field of the water is proportional to r. Thus, the viscous shear $\sigma \propto \eta v/l \propto \eta v/r$. The total drag force is thus $F_d \sim A\sigma \propto \eta r v$, where A is ball's area of contact with the water.

3. Alternatively, to obtain $F_d \propto \eta r v$, make use of the Stokes drag relation $F_d = 6\pi\eta K r v$, where K is a dimensionless constant that takes into account that the ball is not in infinite homogeneous fluid.

Marking scheme:

a) $\omega_0 \propto \sqrt{g/r}$	0.4 pts
stated without justification	-0.2 pts
effective mass $\propto r^3$	0.2 pts
just the mass of the ball considered	-0.1 pts
effective returning force $\propto r^2$	0.1 pts
$\omega_0 \propto \sqrt{g/r}$	0.1 pts
b) $F_d \propto \eta r v$	0.6 pts
no justification	-0.3 pts
Stokes without constant K	-0.1 pts
c) $\beta \propto 1/r^2$	0.3 pts
d) $\frac{\beta^2}{\omega_0^2} = 1 - \frac{\omega^2}{\omega_0^2}$	0.4 pts
e) $\frac{\beta^2}{\omega_0^2} \propto \frac{1}{r^3}$	0.2 pts
f) Answer: 0.271	0.1 pts

Solution 2: The oscillation of the half-sunk sphere is driven by the change in buoyancy force, which is proportional to the change in displaced water volume. Thus, the restoring force $F_r \propto r^2 x$, where x is the displacement of the sphere.

As discussed in Solution 1, the drag force $F_d \propto rv = r\dot{x}$. The effective mass of the oscillation $m \propto r^3$. This leads to the equation of motion

$$k_1 r^2 \ddot{x} + k_2 \dot{x} + k_3 r x = 0,$$

where k_1, k_2 and k_3 are constant. In the case with no viscous drag, k_2 = 0, the motion is at frequency

$$\omega_0 = \sqrt{\frac{k_3}{k_1 r}}.$$

With viscous drag, we can get the frequency ω by substituting trial solution $x = e^{\alpha t}$ and using $\omega = \mathrm{Im}\alpha$. This leads to

$$\omega = \sqrt{\omega_0^2 - \frac{k_2^2}{4k_1^2 r^4}},$$

$$\omega^2 = \omega_0^2\left(1 - \frac{k_2^2}{4k_1^2 r^4 \omega_0^2}\right) = \omega_0^2\left(1 - \frac{k_2^2}{4k_1 k_3 r^3}\right).$$

At $r = r_{min}$, $\omega = 0$, so

$$\frac{k_2^2}{4k_1 k_3} = r_{min}^3,$$

and

$$\omega^2 = \omega_0^2\left(1 - \frac{r_{min}^3}{r_0^3}\right),$$

giving $\frac{r_{min}}{r_0} = 0.271$.

Marking scheme:

a) effective mass $\propto r^3$	0.2 pts
just the mass of the ball considered	-0.1 pts
b) effective returning force $\propto r^2$	0.1 pts
c) $\omega_0 \propto \sqrt{1/r}$	0.1 pts
d) $F_d \propto rv$	0.6 pts
no justification	-0.3 pts
Stokes without constant K	-0.1 pts
e) $k_1 r^2 \ddot{x} + k_2 \dot{x} + k_3 r x = 0$	0.3 pts
f) ω in terms of r and ω_0	0.6 pts
if not expressed in terms of ω_0	-0.3 pts
g) Answer: 0.271	0.1 pts